



Dual Promotion Strategy for Cobalt-Based Fischer–Tropsch Catalysts: Hydrogen Spillover and Structural Stabilization by Ag and B

Halime Kord-Tamandani¹ · Younes Ghalandarzahi² · Ali Akbar Mirzaei¹

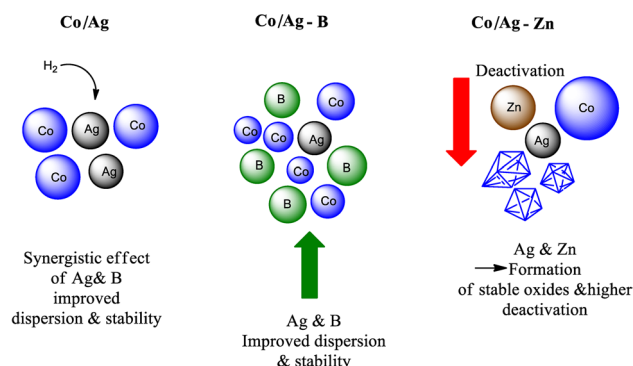
Received: 16 July 2025 / Accepted: 11 November 2025 / Published online: 2 December 2025
 © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

In this study, the deactivation behavior of three granular catalysts—Co/Ag, Co/Ag-B, and Co/Ag-Zn—was investigated in the Fischer–Tropsch synthesis process. Silver was introduced as a primary promoter to enhance cobalt reducibility and facilitate hydrogen spillover. Characterization results from XRD, TPR, FESEM, and EDS analyses revealed that Ag incorporation led to a moderate reduction in cobalt oxide reduction temperature and improved reduction behavior compared to undoped catalysts. However, the promotion effect of Ag alone was limited. The presence of boron significantly improved cobalt dispersion, decreased crystallite size, and reduced the rate of deactivation. In contrast, the addition of Zn resulted in the formation of more stable oxide phases and more difficult reducibility, leading to a noticeable decline in catalytic activity. The Co/Ag-B catalyst, benefiting from the synergistic effect of Ag and B, exhibited the highest structural stability and the lowest deactivation rate. These findings highlight the critical role of rational promoter design in developing next-generation stable cobalt-based catalysts for Fischer–Tropsch synthesis.

Graphical Abstract

Dual-Promotion Effects on Co-based Fischer–Tropsch Catalysts



Keywords Ag · Boron · Catalyst deactivation · Co₃O₄ · Cobalt catalyst · Fischer–Tropsch synthesis · Promoters · Reducibility · Zinc

✉ Halime Kord-Tamandani
 kord.chem96@gmail.com

✉ Ali Akbar Mirzaei
 Mirzaei@hamoon.usb.ac.ir

¹ Department of Chemistry, Faculty of Sciences, University of Sistan and Baluchestan, 98164-161 Zahedan, Iran

² Faculty of Industrial & Mining Khash, University of Sistan and Baluchestan, 98164-161 Zahedan, Iran

1 Introduction

Fischer–Tropsch synthesis (FTS) plays a pivotal role in the catalytic conversion of synthesis gas (a mixture of CO and H₂) into long-chain hydrocarbons, offering a promising alternative to traditional fossil-derived fuels [1]. Among various active metals used in FTS, cobalt-based catalysts are particularly preferred due to their high catalytic activity,

minimal water–gas shift activity, and outstanding selectivity toward linear paraffins [2]. However, despite these merits, the long-term application of cobalt catalysts is significantly hindered by deactivation phenomena, which may arise from sintering, oxidation, carbon accumulation, or strong metal–support interactions [3, 4]. Addressing these deactivation mechanisms is essential for improving catalyst stability and enabling sustained industrial operation of FTS processes.

Alumina (Al_2O_3) is widely used as a catalyst support in cobalt-based Fischer–Tropsch synthesis due to its high surface area, excellent thermal stability, and strong mechanical strength. These properties facilitate the dispersion of cobalt active sites and enhance the structural integrity of the catalyst during prolonged reactions [5, 6].

Promoter incorporation has emerged as an effective strategy to improve both the performance and stability of cobalt catalysts. Silver (Ag), for instance, is known to facilitate hydrogen spillover and decrease the reduction temperature of cobalt oxides, thereby enhancing the formation of active metallic cobalt sites [7–9]. Boron (B), on the other hand, can improve metal dispersion, inhibit sintering, and modify the acid-base properties of the support [10, 11]. Zinc (Zn), meanwhile, can influence the interaction between cobalt and the support and may promote the formation of spinel-type structures, which often result in reduced cobalt reducibility [12]. Although the individual impacts of Ag, B, and Zn have been investigated previously, comprehensive comparative studies evaluating their combined effects on cobalt catalyst deactivation under FTS conditions are scarce. In our previous studies on cobalt-based Fischer–Tropsch catalysts, we have systematically explored the effects of support modification and metal loading on activity and selectivity [13–15]. Building on these findings, the present work focuses on the impact of dual promoters (Ag combined with B or Zn) on catalyst deactivation and structural stability. In the present study, we systematically examine the deactivation behavior of three promoted cobalt catalysts—Co/Ag, Co/Ag-B, and Co/Ag-Zn—using a suite of advanced characterization techniques, including X-ray diffraction (XRD), temperature-programmed reduction (TPR), field emission scanning electron microscopy (FESEM), and energy-dispersive X-ray spectroscopy (EDS). Our goal is to elucidate the links between structural changes, reducibility, and catalytic stability, and to determine the most effective promoter combination for prolonging catalyst lifetime in FTS applications.

2 Experimental

2.1 Catalyst Preparation (Materials)

Three cobalt-based catalysts were synthesized via the incipient wetness impregnation method using $\gamma\text{-Al}_2\text{O}_3$ granules as the support material. Cobalt nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}$) was employed as the cobalt precursor, while silver nitrate (AgNO_3), boric acid (H_3BO_3), and zinc nitrate hexahydrate ($\text{Zn}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}$) were used as sources of silver, boron, and zinc, respectively. The target metal loadings were 15 wt% cobalt, 1 wt% silver, and 0.5 wt% of either boron or zinc, depending on the catalyst formulation. Following each impregnation step, the catalysts were dried at 100 °C for 12 h to remove excess moisture and ensure proper anchoring of the metal precursors onto the support. For the bimetallic catalysts (Co/Ag-B and Co/Ag-Zn), impregnation and drying were carried out sequentially. To stabilize the cobalt oxide phases and minimize sintering or particle agglomeration, an intermediate calcination step was performed at 350 °C for 3 h in static air after each of the first three impregnation steps. Finally, all samples underwent a terminal calcination at 380 °C for 4 h to promote effective dispersion of the active components and the added promoters.

2.2 Catalyst Testing for Fischer–Tropsch Synthesis

Experiments were carried out in a fixed-bed tubular micro-reactor using 1.0 g of calcined catalyst. Prior to reaction, the catalyst was in situ reduced under H_2 (50 mL/min, 2 bar) at 673 K for 12 h. The reactor was then cooled to 413 K and pressurized to 6 bar with H_2/N_2 (1:1) at a space velocity of $4.2 \text{ NL} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$. After stabilization, syngas with $\text{H}_2/\text{CO}=2$ and 5 vol% N_2 (as internal standard) was introduced. The reaction was conducted at 503 K and $5.1 \text{ NL} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$. CO conversion was monitored over 120 h.

Finally, the reactant and product were analyzed on-line using a gas chromatograph (UNICAM PROG C+, TERMO ONIX and USA).

2.3 Catalyst Characterization

The structural and physicochemical properties of the synthesized catalysts were investigated using various characterization techniques. Fourier Transform Infrared (FT-IR) spectra were recorded on a Perkin Elmer Spectrum Two spectrophotometer in the range of $4000\text{--}400 \text{ cm}^{-1}$ using KBr pellets. X-ray diffraction (XRD) patterns of both fresh and calcined samples were obtained with a Bruker D8 Advance diffractometer (Germany) employing $\text{Cu K}\alpha$ radiation (40 kV, 30 mA). Data were collected over a 2θ range of

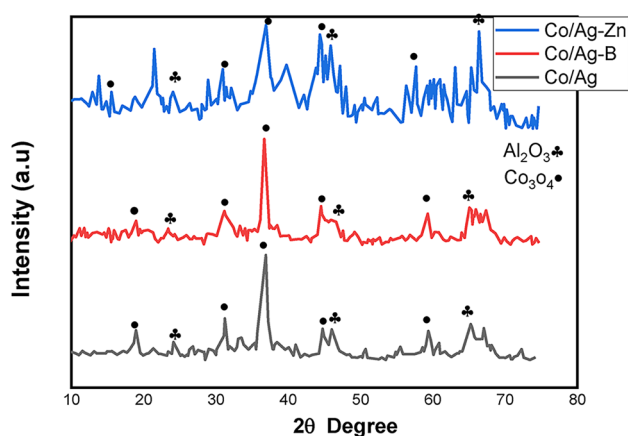


Fig. 1 Diffractogram of cobalt catalysts: Co/Ag, Co/Ag-B and Co/Ag-Zn catalysts

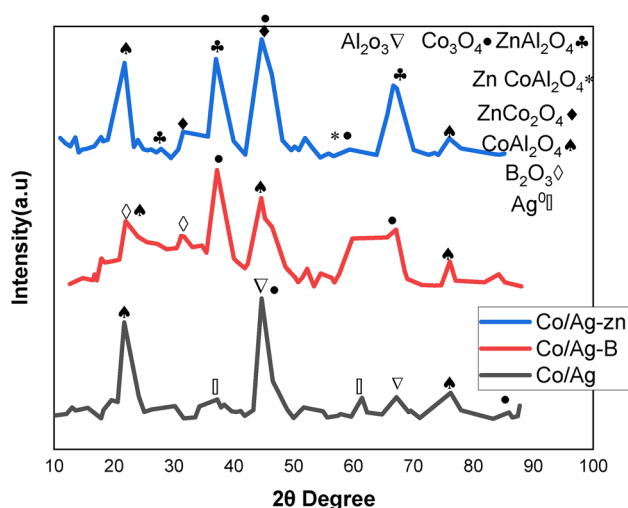


Fig. 2 Diffractogram of cobalt catalysts (Co/Ag, Co/Ag-B, and Co/Ag-Zn) after the reactor tests

5°–80° with a step size of 0.02° and a counting time of 1.0 s. The textural properties, including specific surface area, pore volume, and average pore diameter, were determined by N₂ physisorption using a Quantachrome Nova 2000 automated system (USA). Prior to measurements, all samples were degassed at −196 °C for 66 min. Morphological features and surface elemental distribution were analyzed by field emission scanning electron microscopy (FE-SEM, KYKY Em8000F) equipped with energy dispersive X-ray spectroscopy (EDAX) and elemental mapping (MAP). The maximum resolution of the instrument was 1.2 nm, and samples were sputter-coated with a thin layer of gold before observation. The reducibility of the catalysts was studied by H₂-temperature programmed reduction (H₂-TPR) using a NanoSORD NS91 instrument (Iran). Approximately 30 mg of catalyst was loaded into a U-shaped quartz reactor and heated from room temperature to 950 °C at a rate of 10 °C/min under a reducing gas flow. Elemental composition was

determined by inductively coupled plasma atomic emission spectroscopy (ICP-AES). For sample preparation, 40 mg of catalyst was dissolved in 2 mL of concentrated HNO₃, followed by the addition of 2 mL of 30 wt% H₂O₂. The resulting solution was diluted to 1000 mL with deionized water prior to analysis.

3 Results and Discussion

3.1 XRD Patterns Data

The X-ray diffraction (XRD) patterns of the freshly prepared catalysts Co/Ag, Co/Ag-B, and Co/Ag-Zn are shown in Fig. 1. In addition, the XRD patterns of the catalysts after the reactor tests are presented in Fig. 2 to highlight the structural and phase changes during the reaction. All three samples display prominent diffraction peaks at 2θ values around 31.3°, 36.9°, and 59.5°, which correspond to the (220), (311), and (511) planes of Co₃O₄, confirming the presence of cobalt oxide as the dominant crystalline phase. For the Co/Ag catalyst, the Co₃O₄ peaks are relatively sharp and intense, indicating the formation of larger cobalt oxide crystallites and partial sintering during the calcination process. Additionally, weak peaks observed at approximately 2θ = 38.1° and 44.3° can be attributed to metallic silver (Ag⁰), corresponding to the (111) and (200) planes of face-centered cubic Ag (JCPDS 04-0783).

Although a weak peak from the (222) plane of Co₃O₄ may also be present near 38.1°, its intensity is very low and overlaps with the silver peak, making it difficult to distinguish separately. Therefore, the peak at this position is primarily assigned to metallic silver. In contrast, the Co/Ag-B catalyst shows broader and less intense Co₃O₄ peaks, suggesting smaller crystallite sizes and improved dispersion of cobalt particles. This behavior is likely due to the presence of boron, which inhibits particle growth and enhances phase homogeneity. In contrast, the Co/Ag-Zn sample displays stronger and sharper Co₃O₄ peaks, suggesting larger cobalt oxide crystallites. Additionally, the appearance of slight shoulders near 36–37° may indicate the formation of mixed spinel structures such as ZnCo₂O₄. The presence of these more stable oxide phases may adversely affect the reducibility and catalytic activity. Scherrer analysis of the fresh catalysts indicated average Co₃O₄ crystallite sizes of ~8.8, 10.8 and 17.7 nm for Co/Ag-B, Co/Ag, and Co/Ag-Zn, respectively. These results suggest improved cobalt dispersion in the B-promoted sample and pronounced crystallite growth in the Zn-promoted sample. After the Fischer–Tropsch reaction, the XRD patterns revealed structural changes depending on the promoter: the Co/Ag-B catalyst largely preserved cobalt reflections, whereas Co/Ag showed a partial decrease

in metallic cobalt features along with stronger oxide signals. In contrast, the Co/Ag-Zn catalyst exhibited reflections assignable to Zn–Co–Al spinel phases, indicating that a fraction of cobalt became incorporated into non-reducible structures. These findings are consistent with the TPR results and the observed deactivation behavior. Overall, the XRD analysis demonstrates that boron addition leads to better cobalt dispersion and smaller crystallites, while zinc promotes the formation of more crystalline and possibly less reducible cobalt-containing phases.

3.2 Temperature Programmed Reduction (TPR)

The temperature-programmed reduction (TPR) profiles of Co/Ag, Co/Ag-B, and Co/Ag-Zn catalysts are presented in Fig. 3. All three samples exhibit two major reduction peaks. The first peak, occurring in the range of approximately 280–380 °C, is attributed to the reduction of Co_3O_4 to CoO. The second peak, observed between 450 and 600 °C, corresponds to the further reduction of CoO to metallic cobalt (Co^0) [16–18].

The Co/Ag catalyst shows a moderate reduction profile, with the first peak centered at ~330 °C and the second at ~550 °C. These values indicate the presence of cobalt oxide with moderate crystallinity, in line with the XRD results. The presence of Ag slightly shifts the peaks to lower temperatures, consistent with its role in promoting hydrogen spillover and facilitating cobalt reduction [19].

The Co/Ag-B sample exhibits a significantly improved reduction behavior, with both peaks shifted to lower temperatures (~300 °C and ~500 °C). This shift indicates easier reducibility and suggests that boron improves cobalt dispersion, resulting in smaller oxide particles that reduce more readily. This observation aligns with the broader XRD peaks observed in the same sample.

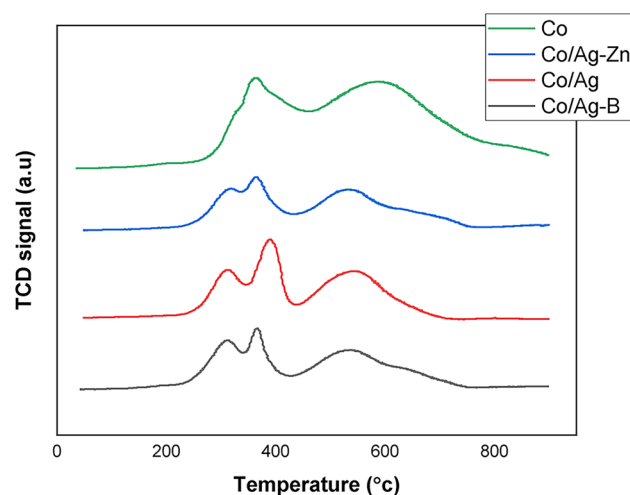


Fig. 3 TPR curves of cobalt catalysts supported on alumina granules

In contrast, the Co/Ag-Zn sample displays higher-temperature reduction peaks (~360 °C and ~570 °C), indicating more difficult reducibility. This behavior is attributed to the formation of more stable oxide or spinel-type structures (e.g., ZnCo_2O_4), which are harder to reduce. This correlates well with the sharper and more intense Co_3O_4 peaks observed in the XRD patterns.

Overall, TPR analysis confirms that boron facilitates cobalt reduction by promoting smaller and more dispersed oxide phases, whereas zinc hinders the reduction process due to the formation of more stable cobalt-containing structures.

3.3 FE-SEM, EDAX and MAP

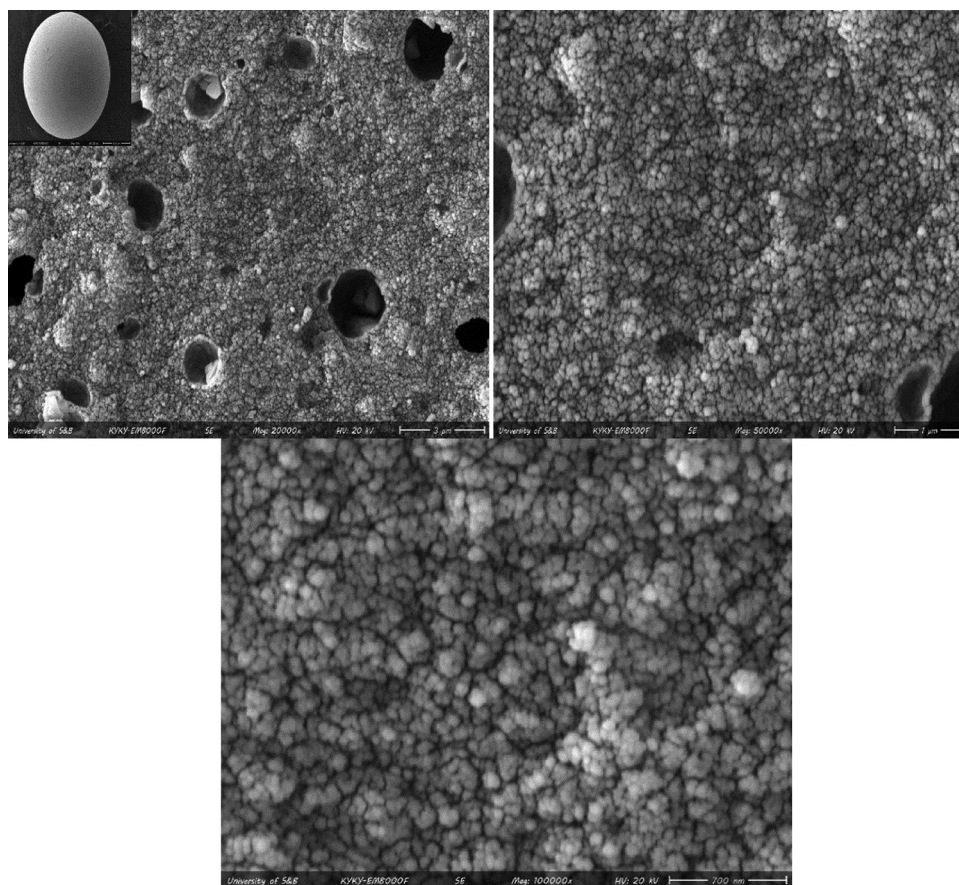
Field emission scanning electron microscopy (FESEM) images and corresponding energy-dispersive X-ray spectroscopy (EDS) data of the Co/Ag, Co/Ag-B, and Co/Ag-Zn catalysts are shown in Figs. 4, 5, 6, 7, 8 and 9; Table 1, respectively. These analyses were used to evaluate the surface morphology and elemental distribution of the catalysts after reaction.

The Co/Ag catalyst exhibits relatively heterogeneous morphology, with aggregated cobalt-rich domains indicating mild sintering. The EDS analysis confirms the presence of Co (4.62 wt%) and a small amount of Ag (0.12 wt%), suggesting partial agglomeration and limited dispersion of active sites. The relatively high oxygen content (76.9 wt%) implies the existence of unreduced or reoxidized cobalt oxides.

The Co/Ag-B sample displays a more homogeneous and porous surface morphology with smaller and well-dispersed cobalt particles. This is in agreement with the broader XRD peaks and lower reduction temperatures observed in TPR. The EDS results show increased Co content (5.56 wt%) and reduced oxygen (73.6 wt%), indicating improved metal loading and better reduction. The presence of B (not directly detected due to its low atomic number) is inferred from structural improvements and improved cobalt distribution.

In contrast, the Co/Ag-Zn sample exhibits an irregular surface with large oxide clusters and less uniform dispersion. EDS results reveal lower Co content (3.67 wt%) and higher oxygen (81.7 wt%), which supports the presence of more stable cobalt oxide species and possible formation of ZnCo_2O_4 -like phases. The presence of Zn (0.09 wt%) and Ag (0.11 wt%) is detected, but their dispersion appears limited. The bulk cobalt and silver contents of the fresh catalysts, determined by ICP and presented in Table 2, indicate similar Co loadings across all catalysts, with Ag contents also comparable. These results confirm successful incorporation of the promoters and consistency of the overall elemental composition with the intended formulations.

Fig. 4 FESEM image of cobalt catalysts (Co/Ag)



In summary, the FESEM and EDS analyses confirm that the Co/Ag-B catalyst benefits from enhanced cobalt dispersion and reduced sintering, while Co/Ag-Zn suffers from phase segregation and more extensive oxidation. These results are consistent with the structural and reduction trends observed in XRD and TPR analyses.

3.4 Fischer–Tropsch Synthesis (FTS)

3.4.1 CO Conversion During Time-on-Stream Test

The CO conversion profiles of the Co/Ag, Co/Ag-B, and Co/Ag-Zn catalysts over 120 h of time on stream are shown in Fig. 10, and the corresponding quantitative data are summarized in Table 3. According to the data in Table 3, the results obtained, all catalysts exhibited a gradual decline in activity over time, consistent with typical deactivation trends observed in cobalt-based Fischer–Tropsch catalysts.

The Co/Ag-B catalyst displayed the highest stability, maintaining over 60% conversion after 120 h. This is attributed to its enhanced cobalt dispersion, smaller crystallite size, and improved reduction behavior, as confirmed by XRD, TPR, and FESEM results.

In contrast, the Co/Ag-Zn catalyst experienced the most significant deactivation, with CO conversion decreasing

from ~60% to ~35%. This sharp decline is linked to the formation of more stable cobalt oxide/spinel phases, as indicated by sharper XRD peaks and high oxygen content in EDS, which limit the availability of active Co^0 sites.

The Co/Ag catalyst exhibited an intermediate behavior, with a moderate decline from ~65% to ~45%. This reflects partial reoxidation and limited cobalt dispersion in the absence of boron.

These results reinforce the structural analyses and highlight the superior deactivation resistance of the Ag–B dual-promoted system under extended reaction conditions.

3.4.2 Catalyst Deactivation Behavior

The CO conversion profiles of the Co/Ag, Co/Ag-B, and Co/Ag-Zn catalysts over 120 h of time on stream are shown in Fig. 10, and the corresponding quantitative data are summarized in Table 3. All catalysts exhibited a gradual decline in activity over time, consistent with typical deactivation trends observed in cobalt-based Fischer–Tropsch catalysts.

The deactivation behavior of the studied catalysts was interpreted based on structural, morphological, and reduction characteristics obtained from XRD, TPR, FESEM, and EDS analyses. The results reveal clear differences in

Fig. 5 Elemental mapping images and EDAX spectrum for catalysts of Co/Ag

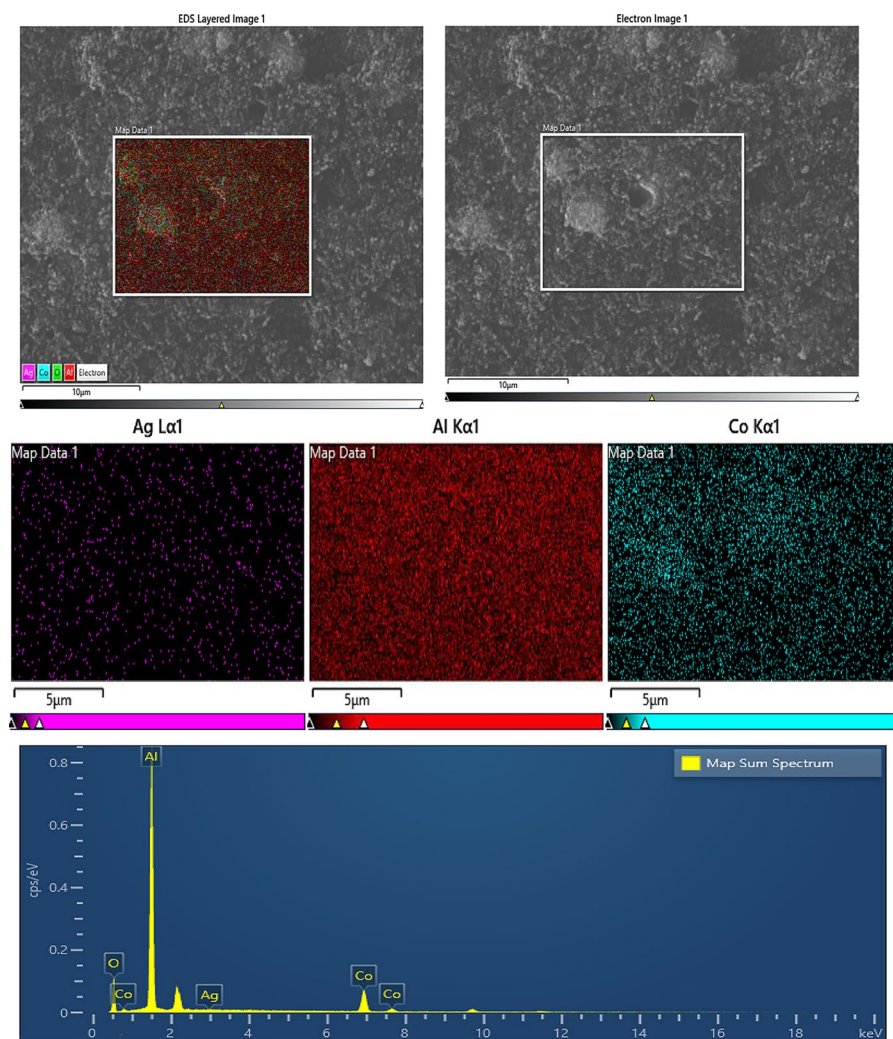


Fig. 6 FESEM image of cobalt catalysts (Co/Ag-B)

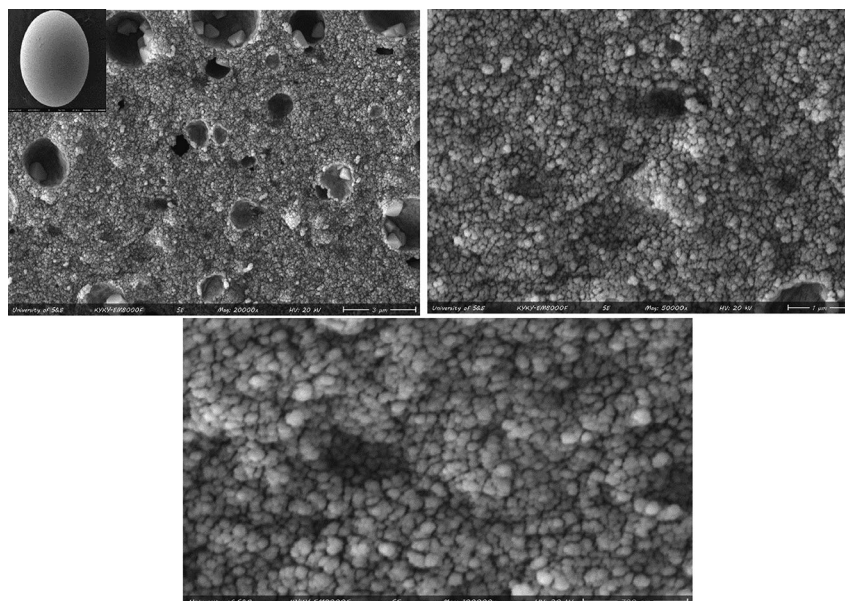


Fig. 7 Elemental mapping images and EDAX spectrum for catalysts of Co/Ag-B

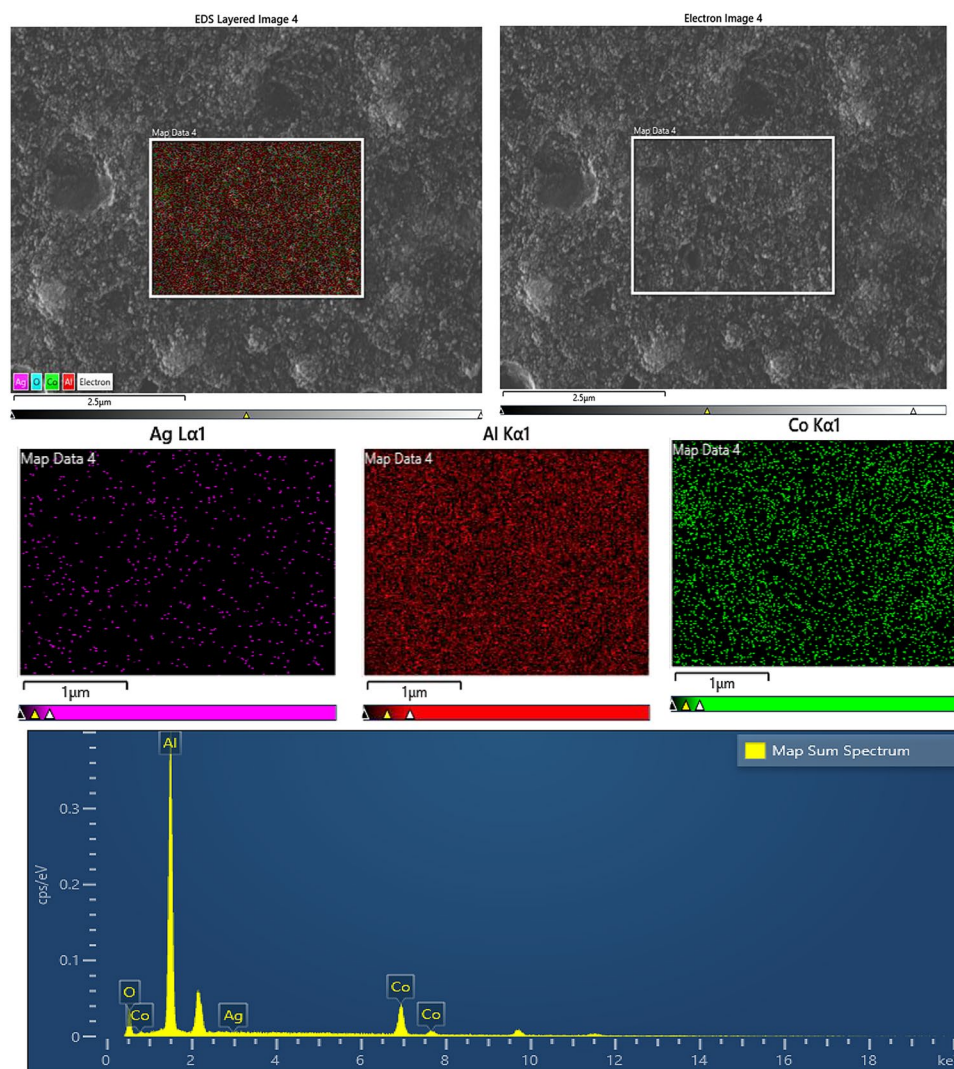


Fig. 8 FESEM image of cobalt catalysts (Co/Ag-Zn)

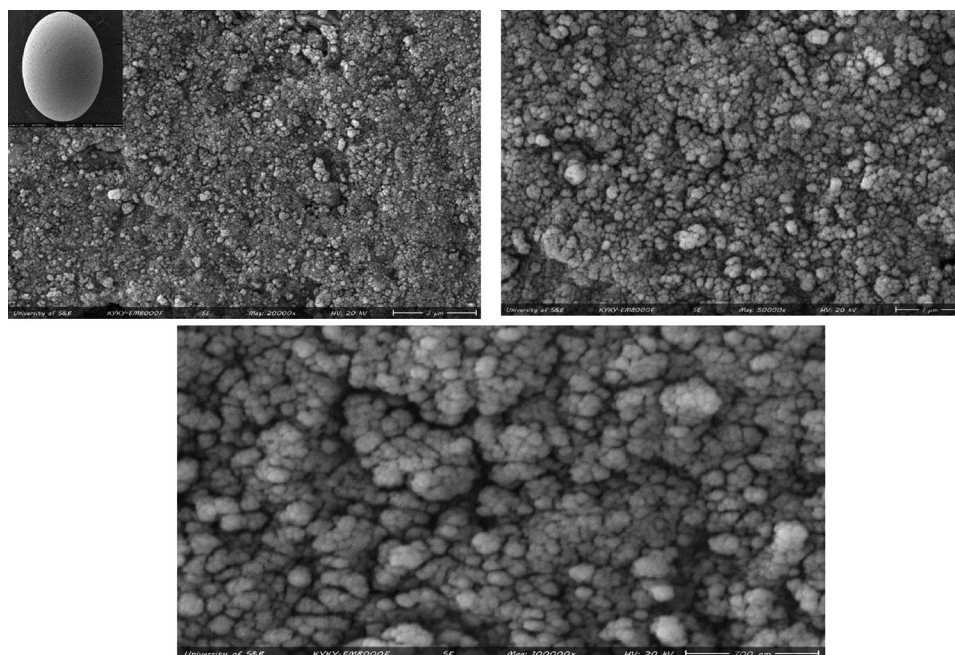


Fig. 9 Elemental mapping images and EDAX spectrum for catalysts of Co/Ag-Zn

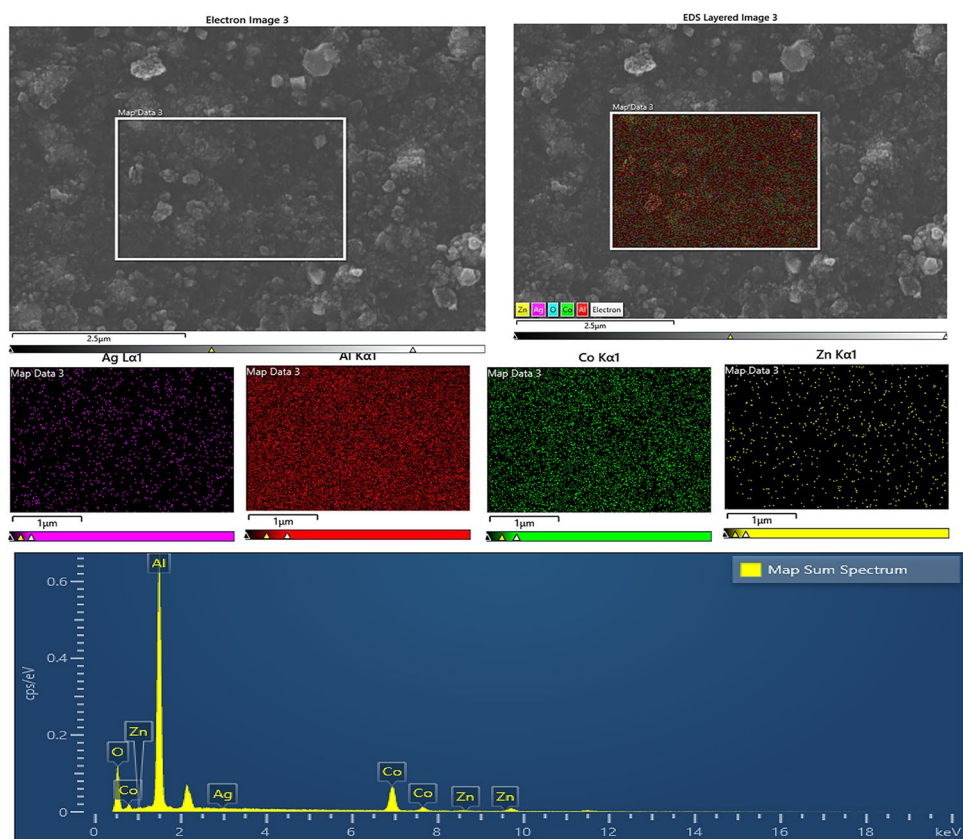


Table 1 EDAX data of the Co/Ag-Zn catalyst, Co/Ag-B catalyst, and Co/Ag catalyst

Element	Co/Ag-Zn	Co/Ag-B	Co/Ag
O	81.75	73.55	76.91
Al	14.38	20.78	18.36
Co	3.67	5.56	4.62
Zn	0.09	—	—
Ag	0.11	0.11	0.12
Total	100.00	100.00	100.00

Table 2 Cobalt and silver content of fresh promoted catalysts determined by ICP (wt%)

Catalyst	Co ICP (wt%)	Ag ICP (wt%)
Co/Ag-B	15.38	0.52
Co/Ag	16.04	0.47
Co/Ag-Zn	15.41	0.44

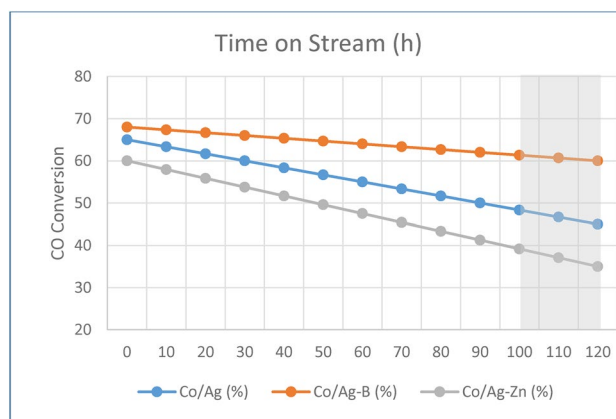


Fig. 10 CO conversion vs. time (0–120 h) for Co/Ag, Co/Ag-B and Co/Ag-Zn. Shaded area = experimental steady-state window (100–120 h). Initial conversion = average of 0–2 h

the extent and mechanism of deactivation depending on the

Table 3 Performances of the catalysts in a fixed bed reactors

Sample	CH ₄ Selectivity (%)	C ₂ –C ₄ Selectivity (%)	C ₅ ⁺ Selectivity (%)	CO Initial Conversion (%)	CO Steady-State Conversion (%)	Activity Loss ^a (%)	CTY ^b (10 ^{−5} mol CO·gCo ^{−1} ·s ^{−1})
Co/Ag	15.5	22.5	64	66.0	45.5	31.06	6.3
Co/Ag-B	11.2	20.5	68.9	69.0	60.5	12.31	7.2
Co/Ag-Zn	18.6	26	56.8	61.0	36	40.98	5.5

^aActivity loss were calculated as (CO initial conversion – CO steady state conversion)/CO initial conversion × 100

^bCobalt time yield and hydrocarbon selectivity were collected at 110–120 h

nature of the promoter.

The Co/Ag catalyst showed moderate crystallinity (XRD), incomplete reduction of cobalt oxides (TPR), and signs of mild sintering (FESEM), along with limited cobalt content and high oxygen levels (EDS). These observations indicate partial deactivation due to moderate dispersion and reoxidation of cobalt [3, 20].

The Co/Ag-B catalyst demonstrated the highest structural and reduction stability. Broad XRD peaks, lower reduction temperatures, homogeneous morphology, and higher cobalt content all point to enhanced cobalt dispersion and smaller crystallite sizes. Boron likely retards sintering and promotes a more stable active phase, as reported in previous studies [10]. As a result, this catalyst exhibited the lowest tendency toward deactivation under FTS conditions.

In contrast, the Co/Ag-Zn catalyst suffered from the most severe deactivation. It exhibited intense and sharp Co_3O_4 peaks (XRD), higher reduction temperatures (TPR), irregular surface morphology (FESEM), and lower cobalt content with the highest oxygen content (EDS). These features suggest that zinc promotes the formation of more stable cobalt oxide or spinel phases (e.g., ZnCo_2O_4), which hinder reduction and limit cobalt accessibility, leading to significant activity loss [21, 22].

In summary, the promoter effect on deactivation resistance followed the trend:

$\text{Ag} - \text{B} > \text{Ag} > \text{Ag} - \text{Zn}$,

with the Co/Ag-B catalyst demonstrating superior structural integrity, cobalt reducibility, and dispersion, making it the most stable catalyst among the three. These structural findings are in excellent agreement with the catalytic performance data presented in Fig. 10; Table 3.

4 Conclusion

This study provided a comprehensive assessment of the deactivation behavior of three cobalt-based catalysts—Co/Ag, Co/Ag-B, and Co/Ag-Zn—employed in the Fischer–Tropsch synthesis process. Structural and reduction analyses revealed that while silver (Ag) served as a moderate promoter, its effect alone was limited. However, the addition of boron in Co/Ag-B introduced a clear synergistic effect. The dual-promotion system of Ag and B significantly improved cobalt dispersion, suppressed crystallite growth, and facilitated reduction, resulting in lower deactivation and better structural integrity.

Conversely, the Ag-Zn dual system in Co/Ag-Zn promoted the formation of spinel-like and stable oxide

structures, limiting the availability of active cobalt sites and leading to higher deactivation.

These findings demonstrate that dual promotion can be a powerful strategy to tailor cobalt catalyst performance, but the choice of co-promoter is critical.

The combination of Ag and B provided the most stable performance under the tested conditions, highlighting the beneficial interplay between hydrogen spillover facilitation (Ag) and cobalt stabilization (B).

Overall, this study underscores the importance of rational dual-promoter design to simultaneously optim.

Acknowledgements The authors gratefully acknowledge financial support from the University of Sistan and Baluchestan for providing the facilities to perform the research. ize dispersion, reducibility, and durability in cobalt-based Fischer–Tropsch catalysts.

Author Contributions Halime Kord-Tamandani conceived and designed the study. Younes Ghalandarzahi performed the catalyst synthesis and characterization. Ali Akbar Mirzaei analyzed the data and contributed to manuscript preparation. All authors discussed the results, reviewed, and approved the final manuscript.

Funding The authors have not disclosed any funding.

Data Availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

References

1. Dry ME (2002) The fischer–tropsch process: 1950–2000. *Catal Today* 71(3–4):227–241
2. Iglesia E (1997) Design, synthesis, and use of cobalt-based Fischer–Tropsch synthesis catalysts. *Appl Catal A* 161(1–2):59–78
3. Bartholomew CH (2001) Mechanisms of catalyst deactivation. *Appl Catal A* 212(1–2):17–60
4. Rytter E, Holmen A (2015) Deactivation and regeneration of commercial type Fischer–Tropsch Co-catalysts—a mini-review. *Catalysts* 5(2):478–499
5. Rytter E, Borg Ø, Enger BC, Holmen A (2019) α -alumina as catalyst support in Co Fischer–Tropsch synthesis and the effect of added water; encompassing transient effects. *J Catal* 373:13–24
6. Gholami Z, Tišler Z, Svobodová E, Hradecká I, Sharkov N, Gholami F (2022) Catalytic performance of Alumina-supported Cobalt carbide catalysts for Low-Temperature Fischer–Tropsch synthesis. *Catalysts* 12(10):1222
7. Chen W, Pestman R, Chiang F-K, Hensen EJ (2018) Silver addition to a Cobalt Fischer–Tropsch catalyst. *J Catal* 366:107–114

8. Jacobs G, Ribeiro MC, Ma W, Ji Y, Khalid S, Sumodjo PT, Davis BH (2009) Group 11 (Cu, Ag, Au) promotion of 15% Co/Al₂O₃ Fischer–Tropsch synthesis catalysts. *Appl Catal A* 361(1–2):137–151
9. Jacobs G, Ma W, Davis BH (2014) Influence of reduction promoters on stability of cobalt/ γ -alumina Fischer–Tropsch synthesis catalysts. *Catalysts* 4(1):49–76
10. Saeys M, Tan KF, Chang J, Borgna A (2010) Improving the stability of Cobalt Fischer–Tropsch catalysts by Boron promotion. *Ind Eng Chem Res* 49(21):11098–11100
11. Tan KF, Chang J, Borgna A, Saeys M (2011) Effect of Boron promotion on the stability of Cobalt Fischer–Tropsch catalysts. *J Catal* 280(1):50–59
12. Gnanamani MK, Hamdeh HH, Jacobs G, Qian D, Liu F, Hopps SD, Thomas GA, Shafer WD, Xiao Q, Hu Y (2016) Fischer–Tropsch synthesis: effect of Cu, Mn and Zn addition on activity and product selectivity of Cobalt ferrite. *RSC Adv* 6(67):62356–62367
13. Rezazadeh E, Mirzaei AA, Atashi H (2021) Morphology engineering of γ -alumina microgranules as support of Cobalt catalysts used for Fischer–Tropsch synthesis: an effective strategy for improving catalytic performance. *J Phys Chem C* 125(32):17718–17733
14. Kord-Tamandani H, Mirzaei AA, Ghalandarzahi Y, Zohdi SH (2024) Effects of surfactant and promoters on the catalytic performance of Co-catalysts supported on γ -Al₂O₃ granules for Fischer–Tropsch synthesis. *S Afr J Chem Eng* 48:156–166
15. Kord-Tamandani H, Mirzaei AA, Zohdi SH (2023) Significant improvement of catalytic performance for the Fischer–Tropsch synthesis over the Cobalt catalysts supported on γ -Al₂O₃ granules: simultaneous effects of glucose surfactant and promoter. *Chem Pap* 77(12):7653–7668
16. Khodakov AY, Griboval-Constant A, Bechara R, Villain F (2001) Pore-size control of Cobalt dispersion and reducibility in mesoporous silicas. *J Phys Chem B* 105(40):9805–9811
17. Prieto G, Martínez A, Concepción P, Moreno-Tost R (2009) Cobalt particle size effects in Fischer–Tropsch synthesis: structural and in situ spectroscopic characterisation on reverse micelle-synthesised Co/ITQ-2 model catalysts. *J Catal* 266(1):129–144
18. Jacobs G, Das TK, Zhang Y, Li J, Racouillet G, Davis BH (2002) Fischer–Tropsch synthesis: support, loading, and promoter effects on the reducibility of Cobalt catalysts. *Appl Catal A* 233(1–2):263–281. [https://doi.org/10.1016/S0926-860X\(02\)00195-3](https://doi.org/10.1016/S0926-860X(02)00195-3)
19. Wang Z, Pokhrel S, Chen M, Hunger M, Mädler L, Huang J (2013) Palladium-doped silica–alumina catalysts obtained from double-flame FSP for chemoselective hydrogenation of the model aromatic ketone acetophenone. *J Catal* 302:10–19
20. Rytter E, Tsakoumis NE, Holmen A (2016) On the selectivity to higher hydrocarbons in Co-based Fischer–Tropsch synthesis. *Catal Today* 261:3–16
21. Gawande KB, Gawande SB, Thakare SR, Mate VR, Kadam SR, Kale BB, Kulkarni MV (2015) Effect of zinc: Cobalt composition in ZnCo₂O₄ spinels for highly selective liquefied petroleum gas sensing at low and high temperatures. *RSC Adv* 5(50):40429–40436
22. Du J, Yan J, Hong J, Zhang Y, Chen S, Li J (2015) Catalytic performance of Co/Zn–Al₂O₃ Fischer–Tropsch catalysts: a comparative study of zinc introduction methodologies. *RSC Adv* 5(74):60534–60540

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.